

RIDHI SRIKANTH

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SUMMARY

Computer Engineering student with systems experience spanning runtimes, operating systems, and networking. Experienced building and debugging low-level software in C/C++, Python, and Shell, with interests in usable abstractions and reliable infrastructure.

EDUCATION

University of California, San Diego | BS, Computer Engineering | GPA: 3.975

Graduation: June 2027

BS/MS Applicant (Expected MS Graduation: June 2028)

- **Selected Coursework:** Operating Systems, Networks, Parallel Computing, Computer Architecture, Algorithms
- **Awards and Honors:** CSE Dept. Honors Candidate, HKN IEEE Honor Society; Provost Honors (9 consecutive quarters)
- **Selected Programs:** Brown Systems Week; Cornell–Maryland–Max Planck Research School (CMMRS) 2026

EXPERIENCE

Research Lead | SALAD Lab UCSD | Advisor: Professor Michael Coblenz

A Python–Shell Hybrid Programming Model

June 2025 - Present

- Extended Python with native shell constructs, enabling Python programs to directly execute shell commands and compose system-level workflows through pipelines and I/O redirection
- Extended CPython across parsing, AST construction, bytecode generation, and C-level evaluation to implement the new language features and execution model
- Engineered interoperability between Python and shell processes, including persistent shell execution and automatic conversion between Python and shell data representations
- Diagnosed and resolved compiler and runtime failures using GDB, LLDB, AddressSanitizer, and low-level execution tracing
- Validated the extended interpreter against 33K+ CPython tests, debugging runtime regressions
- Built a shell-to-extended-Python translation system to evaluate expressiveness against real world shell scripts
- Authored a paper accepted to SPLASH SRC 2026; currently authoring an extended research manuscript targeting publication

Research Assistant | ATLAS Lab Brown University | Advisor: Professor Nikos Vasilakis

Stream Monitor Injection

July 2026 - Present

- Developing runtime monitoring infrastructure to detect silent failures in shell pipelines using static and dynamic analysis.
- Contributed to RT, a type system for shell programs, with new type rules, inference heuristics, and richer type abstractions
- Building a reproducible benchmark of real-world shell pipeline failures from Stack Overflow and GitHub, implementing affected shell scripts and hand-written runtime monitors to reproduce and detect each failure

Full-stack Developer – ElectroTriton With Professor Curt Schurgers | Electrotriton

Aug 2025 – Present

- Developing features for a Django/MySQL homework management platform used across engineering courses
- Implemented automated course cloning and roster synchronization, reducing administrative processing time by ~71% and ~78%, respectively
- Built a new analytics dashboard for instructors to track assignment metrics, including view, start, and completion rates

Research Assistant | SALAD Lab UCSD

A Study of Scientific Computational Notebook Quality

Feb 2025 - June 2025

- Evaluated target scientific repositories from a 500+ dataset to diagnose runtime errors, dependency mismatches, and failures
- Investigated execution failures across diverse software environments and contributed findings to a research manuscript on scientific workflow reproducibility: [link to pre-print](#)

ECE 35 (Analog Design) Course Tutor

Sep 2025 - April 2026

- Managed support for 400+ students, providing tutoring and hands-on lab guidance on real device circuit implementation
- Collaborated in weekly meetings to refine instructional strategies and was responsible for grading quizzes and final exam questions on fundamental circuit theory concepts.

SKILLS

Programming Languages: OpenCL, Python, C, C++, TypeScript/JavaScript, SQL, Shell, Java

Systems & Tools: Linux, GDB, LLDB, AddressSanitizer, Git, Docker, CI/CD

Technologies: React, Flask, Django, Firebase, Agile

LEADERSHIP

- **IEEE UCSD Vice Chair Events (2026–27):** Lead the events team in organizing technical, professional, outreach, and social events for the UC San Diego IEEE community, partnering with industry and campus organizations.
- **IEEE UCSD Technical Chair (2024–26):** Led 5+ technical workshops for 100+ students, including live-coding sessions on React and recommendation systems